AHRI Call Dashboard Project

Turning inbound & outbound call records into a Power BI dashboard

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Background

The **Australian HR Institute (AHRI)** supports HR professionals through membership, certification, training and events.

AHRI hears from members every day

Members get in touch three ways:

Channel	Reporting today
Email	CRM dashboard
Online forms	CRM dashboard
Phone (Microsoft Teams)	**Raw Excel export only**

~10,000 phone interactions in five months, with **no dashboard**

3,986 inbound calls + 6,046 outbound attempts, 6 Apr – 1 Sep 2026

Business problem

AHRI has no visibility and needs a clearer way to understand of

- **When** do members call?
- **Where** do their calls go?
- **How** do calls end, and how long do they take?
- **Who** is calling members back, and when?

Our goal

Turn the Teams phone exports into a Power BI dashboard that shows AHRI what's happening on the phones.

In scope: inbound and outbound activity across the period of the export. **Out of scope:** live reporting and individual staff performance.

Business questions

	Question	Dashboard page
Inbound	How many calls arrive, and when?	Inbound Call Overview
Inbound	Where do callers go?	Routing and queue destinations
Inbound	How do calls end, and how long do they take?	Outcome and Duration
Outbound	Who is making outbound calls, and when?	Outbound activity
Outbound	How do outbound calls end, and how long do they last?	Outbound outcome and duration

Data

Inbound

- Teams Direct Routing export
- 6 April – 1 September 2026
- **3,986 unique calls** through main IVR

Outbound

- Export Outbound file
- 6 April – 1 September 2026
- Focus on **18 staff** mapped to three teams: **6,046 attempts**

IDA and EDA: checking and exploring the data

IDA · Check and prepare

Initial data analysis · Power Query

- **Inbound quality:** 334 missing account names and 785 missing start/end timestamps. Retained the rows with unknown values.

- **Scope:** grouped inbound rows by shared ID, selected Main IVR groups and filtered outbound to 18 approved staff.
- **Consistency:** cleaned account labels, set data types and converted UTC timestamps to Melbourne time.

EDA · Explore patterns

Exploratory data analysis · Power BI and DAX

- **Timing:** compared daily, weekday, hourly and monthly activity, including a 7-day trailing average.
- **Routing and teams:** compared queue/overflow destinations and outbound volumes across teams.
- **Outcomes and duration:** examined technical statuses and duration bands. Compared mean and median to assess the effect of long calls.

Scope check: 9,859 inbound records form 6,361 shared-ID groups. Our inbound analysis focuses on the **3,986 containing Main IVR**.

Weekday averages include zero-activity days and exclude boundary dates. Full-month comparisons cover May–August.

IDA means initial data analysis. We checked what each record represents, missing values, duplicates and timestamp coverage before calculating the dashboard measures.

We retained records with missing account names or timestamps because their other fields could still be useful. We did not invent replacement values. No exact duplicate inbound rows were found. Grouping related records by shared ID is therefore different from removing duplicates.

The 9,859 inbound records form 6,361 shared-ID groups, of which 3,986 contain Main IVR. Vanessa confirmed that groups without Main IVR are mostly direct calls to staff. This supported our decision to focus the inbound analysis on Main IVR. These groups are not counts of unique people.

EDA means exploratory data analysis. We used Power BI charts and DAX measures to investigate when activity occurs, which destinations and teams account for it, and how technical outcomes and recorded durations vary.

For fairer comparisons, weekday averages include zero-activity days and exclude the first and last observed dates. Public holidays remain included, and continuous coverage between the boundary dates is assumed. The trailing average covers the current day and the previous six calendar days.

These are descriptive findings. Technical success does not prove that a member spoke to staff, and recorded duration does not establish queue waiting time.

Approach

From two Excel exports to five dashboard pages

1

CleanTeams Direct Routing + Outbound exports; fix timestamps to Melbourne time

2

GroupUsed shared IDs to identify related call records: 6,361 groups, **3,986 via Main IVR**

3

ClassifyMain IVR, primary queue or overflow; outbound call staff mapped to 3 teams

4

MeasureDAX measures for volume, routing, outcomes and duration

5

Build5 Power BI pages: 3 inbound, 2 outbound

Inbound: 6 Apr – 1 Sep 2026, calls through the Main IVR. Outbound: same period, 18 staff in Membership, HRSC and Events.

Inbound Results

Calls follow office hours, and Tuesday is the busiest day

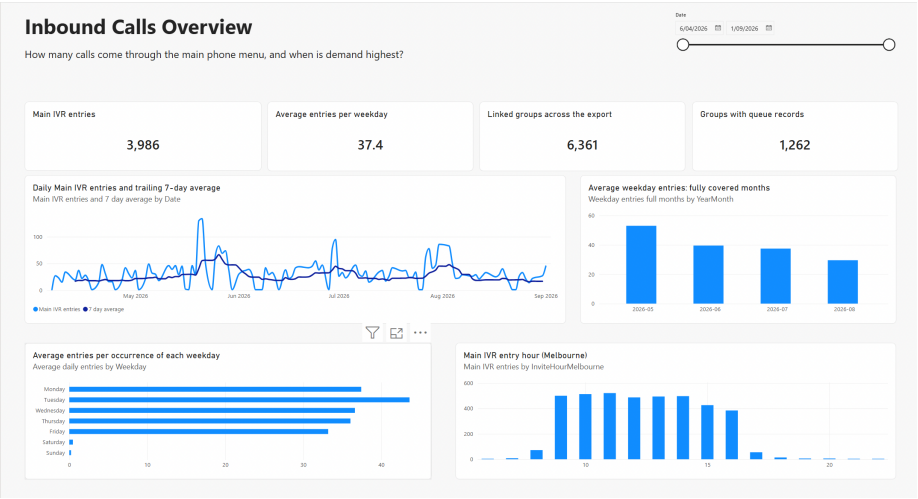


Figure 1: Page 1 · Inbound calls overview

- **3,986** calls, about **37 per weekday**
- Steady **9am–5pm**; almost nothing outside it
- **Tuesday** is busiest (~44/day), Friday quietest (~33)
- Weekday average fell from ~**53 in May to ~30 in August**

Career Support overflows most

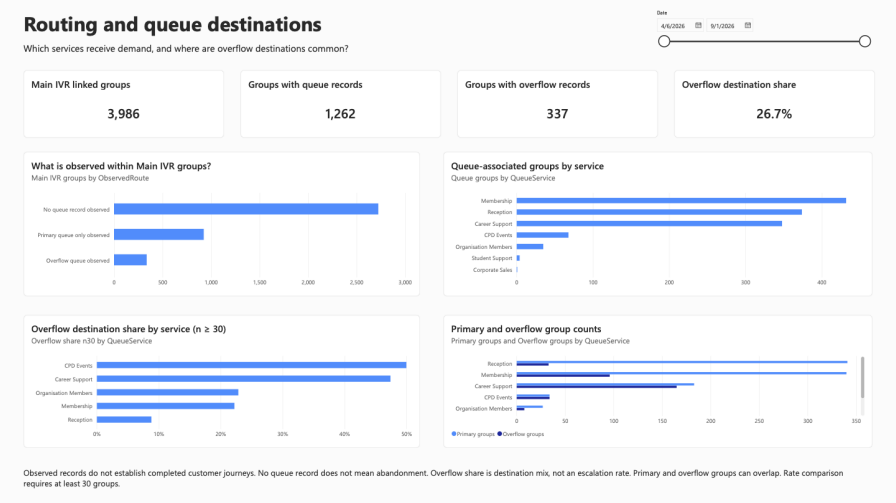


Figure 2: Page 2 · Routing and queue destinations

- **68%** of calls show no queue record (they ended in the menu or went elsewhere)
- **Membership, Reception and Career Support** take most queue demand
- **Career Support:** ~47% of its queue calls reach overflow
- **Reception:** under 10%

Most calls complete; dropped calls tend to be the long ones

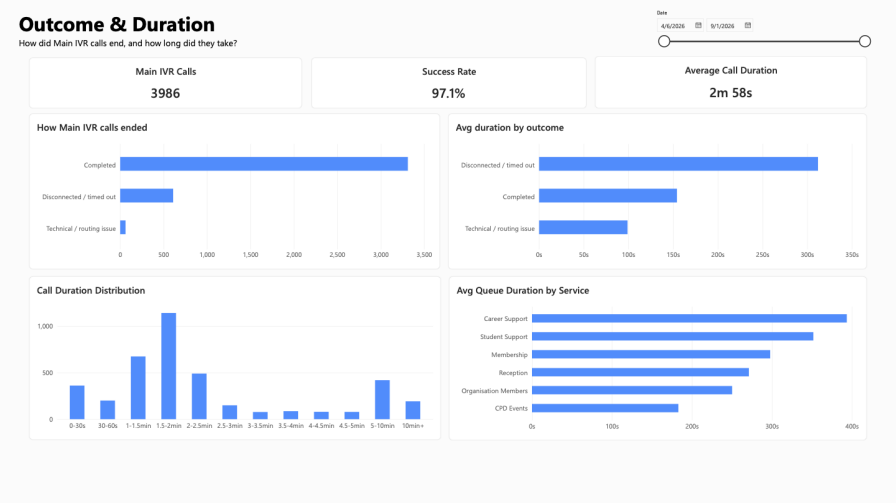


Figure 3: Page 3 · Outcome and duration

- ~**3,300** completed, ~**600** disconnected / timed out
- Typical call: **1.5–2 min**; average **2m 58s**
- Disconnected calls average ~**5 min**, about twice as long as completed ones
- **Career Support** has the longest queue time (~**6.5 min**)

Outbound results

The Membership team makes almost 9 in 10 outbound calls

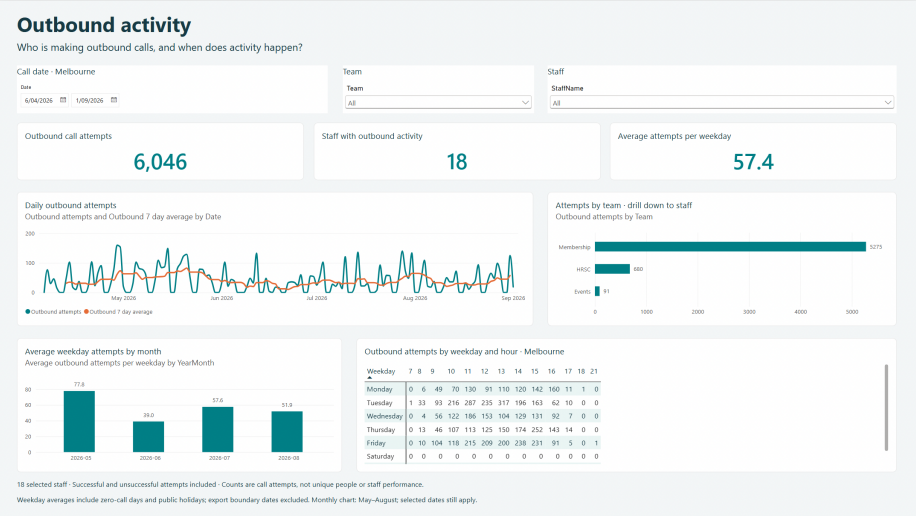


Figure 4: Page 4 · Outbound activity

- **6,046** attempts by **18 staff**, about **57 per weekday**
- **Membership: 87%** (5,275); HRSC 680; Events 91
- Busiest **Tuesday 11am–2pm**; quietest day **Monday**
- May was the peak month (**77.8/weekday**)

91% of outbound calls connect, but most last under a minute

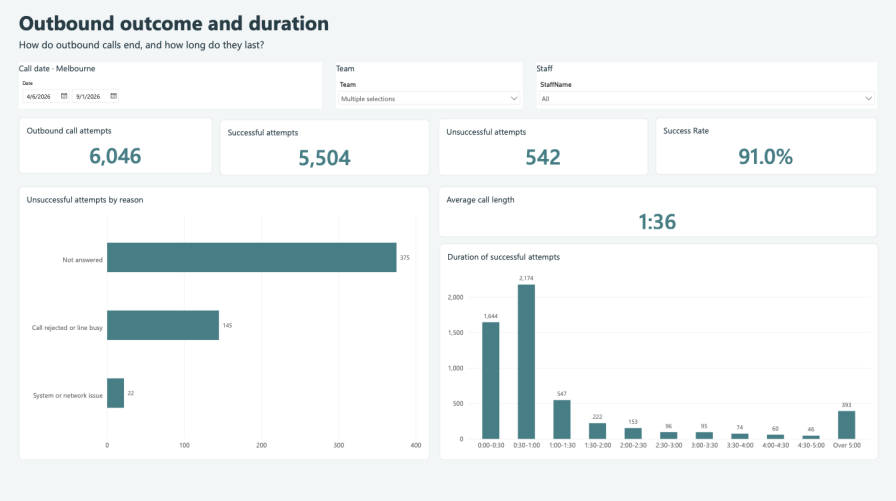


Figure 5: Page 5 · Outbound outcome and duration

- **5,504** connected (**91%**); **542** didn't
- Main failure: **not answered** (375)
- **69%** of connected calls last **under 1 minute**; average **1:36**
- Short calls may be **voicemail**: the export can't tell us

Challenge & Limitations

Inbound

- Whether a staff member answered
- Whether a caller abandoned in the queue
- Whether the caller's question was resolved

Outbound

- Whether the **member** answered, rather than voicemail
- The **purpose** of the call (fees, welcome, other)
- Whether the call achieved its goal